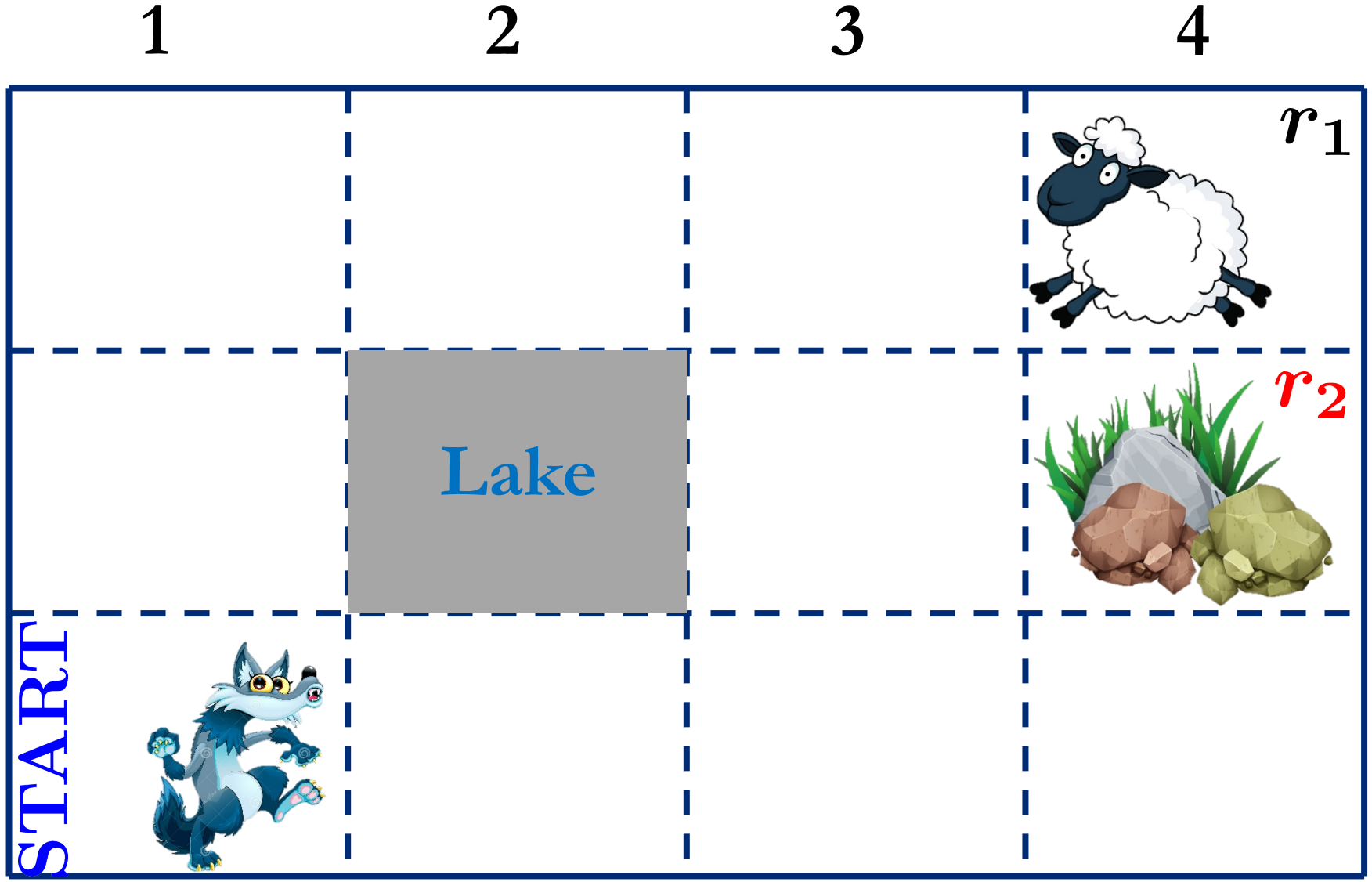
A Game: State of the System

1. System State: s, the location, coordinate of the cell

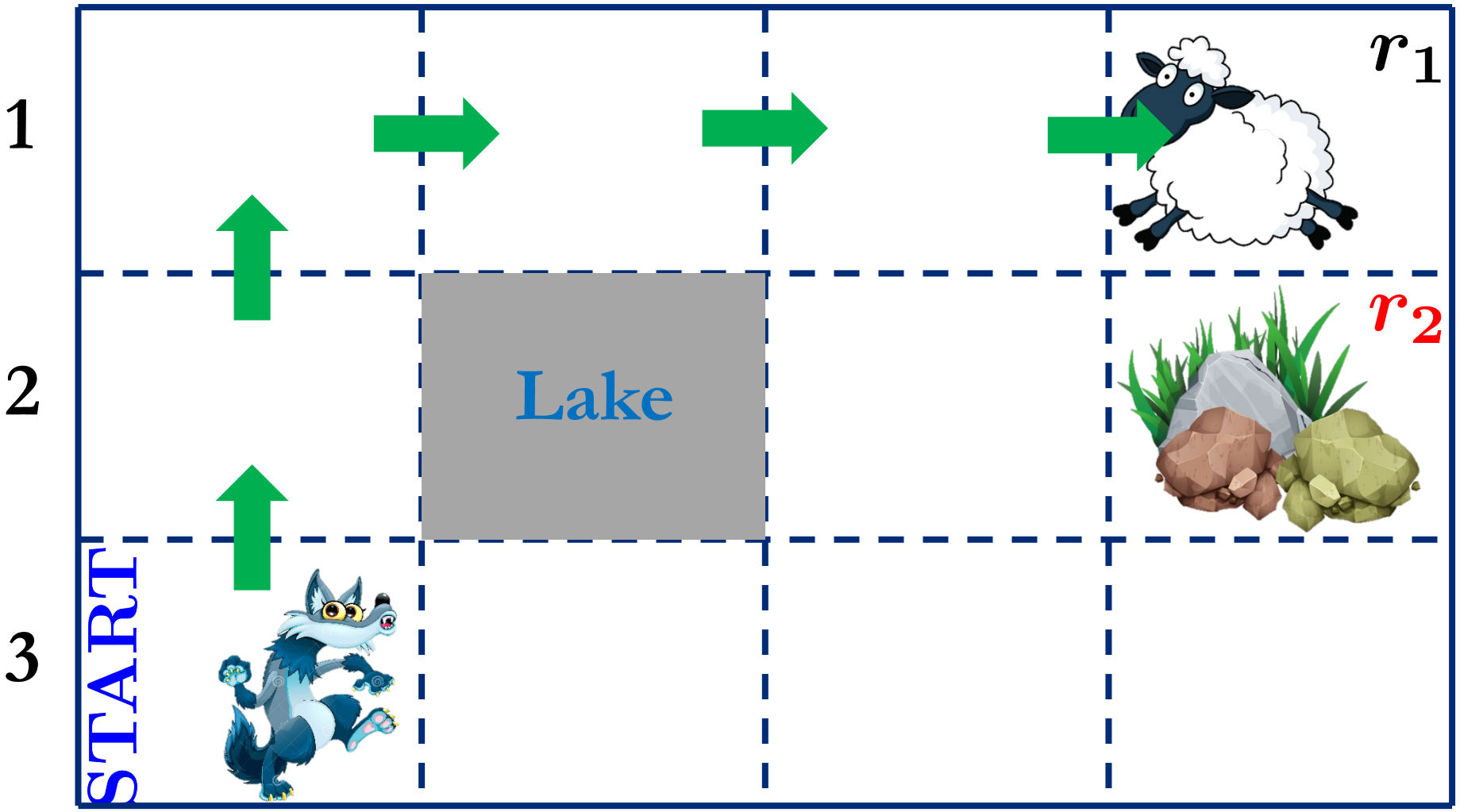


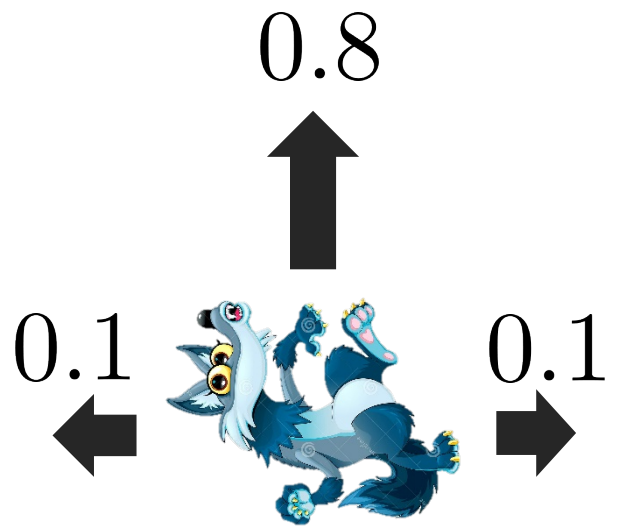
The game ends if the wolf either gets the sheep or hits the rock

The system is fully observable

A Game: Action at Each State

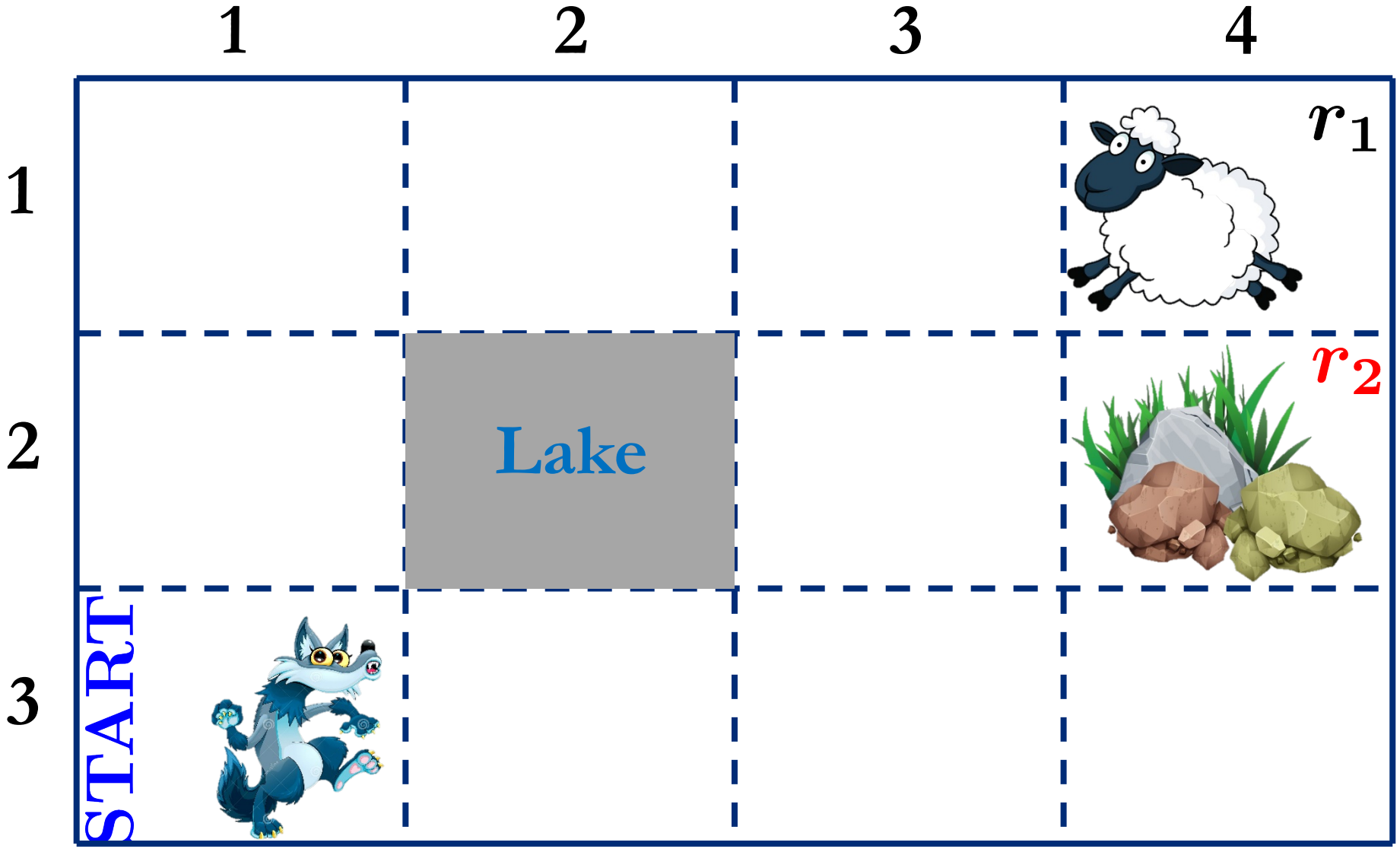
2. $\mathcal{A}(s) = \{\text{Up}, \text{Down}, \text{Left}, \text{Right}\}$: Action at state (s)

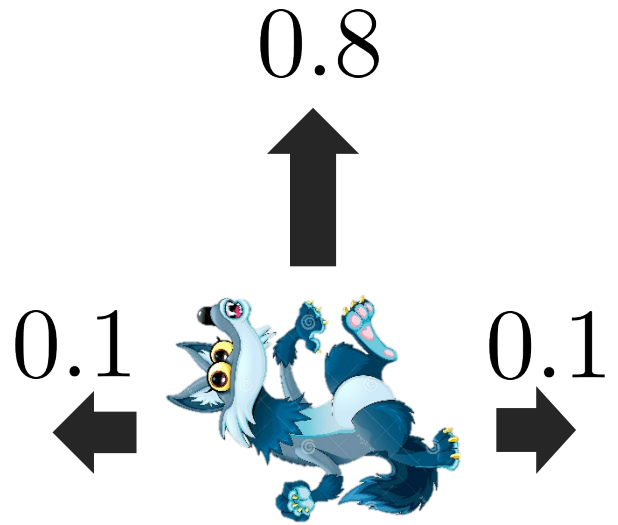




A Game: State Transition Probabilities

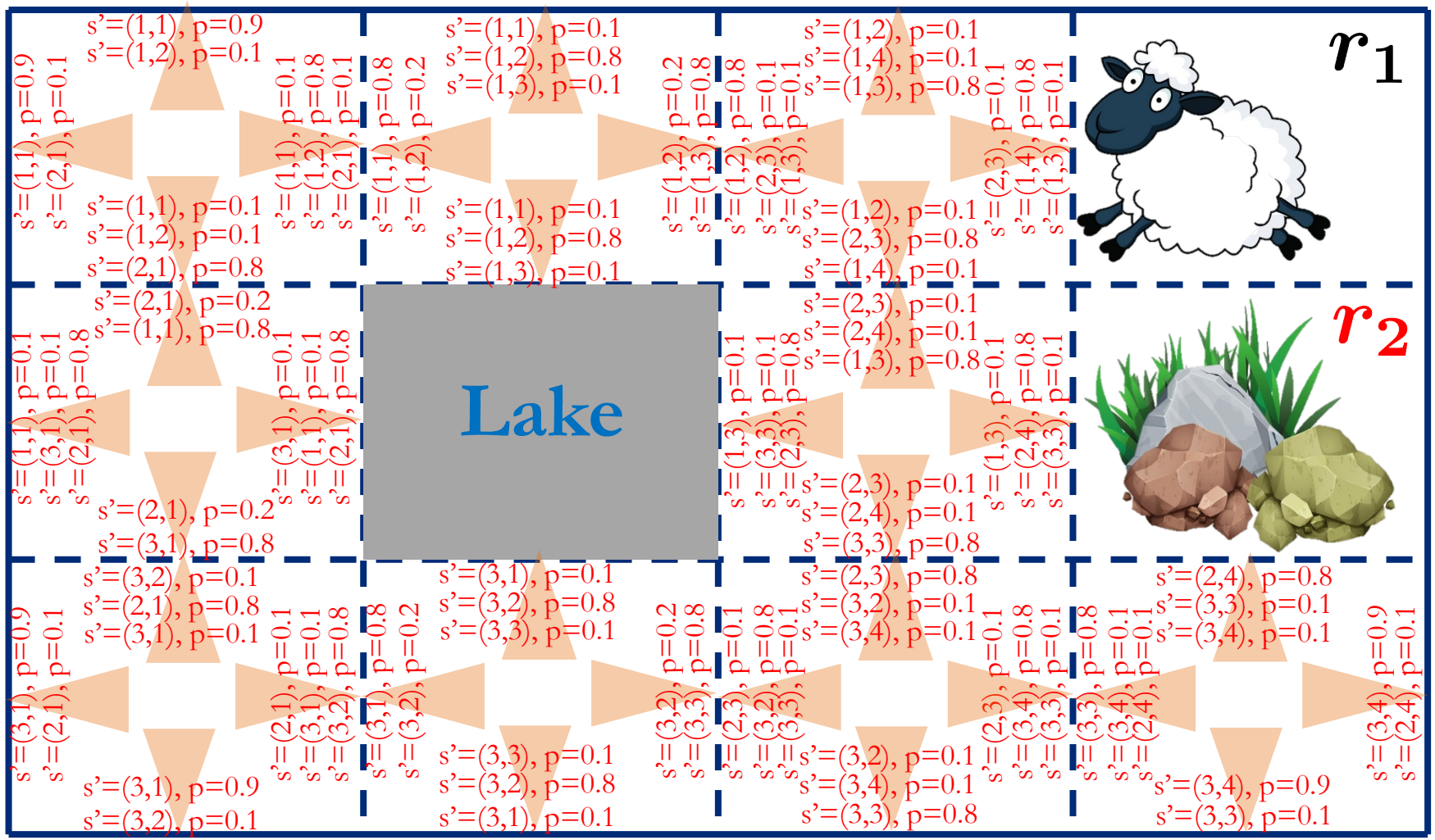
3. $p = \mathbb{P}\{s'|s, a\}$: transition probability

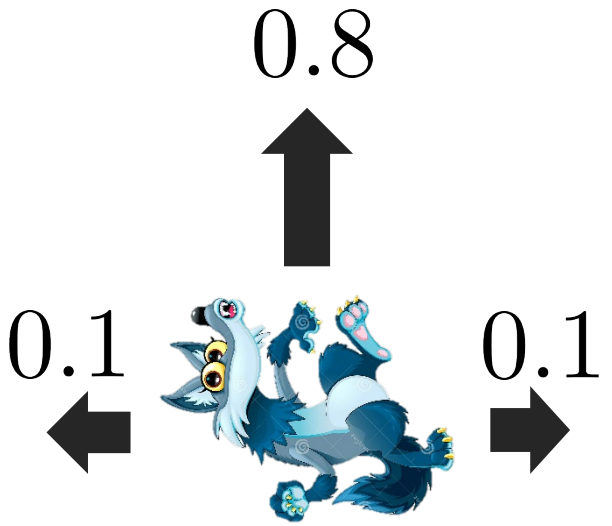




A Game: State Transition Probabilities

3. $p = \mathbb{P}\{s'|s, a\}$: transition probability



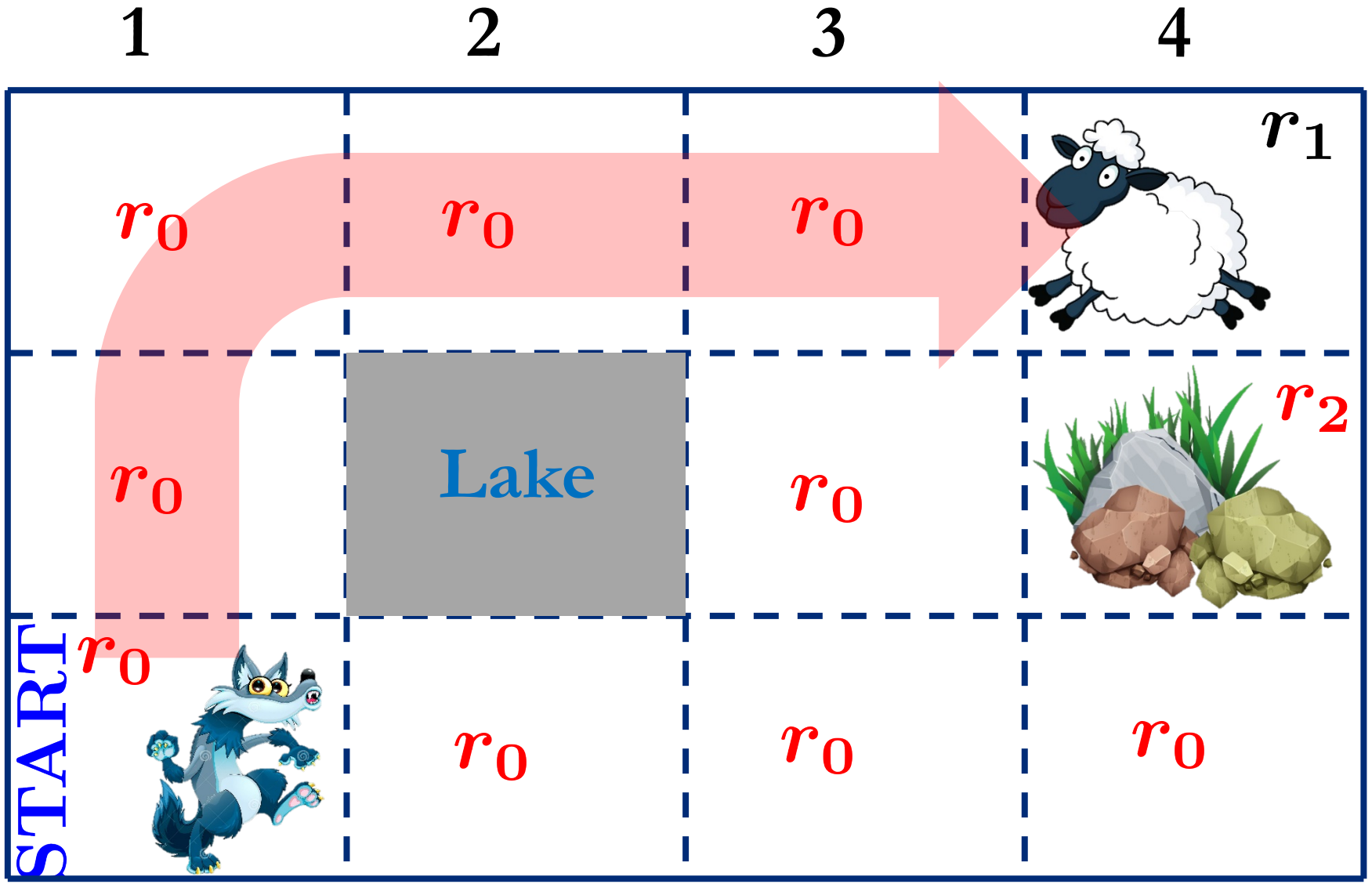


$\mathbb{P}\{(3,2)|(3,1), Up\} = 0.1$

$\mathbb{P}\{(3,1)|(3,1), Left\} = 0.9$

A Game: Reward Function

4. $R(s, a)$: Reward (or Cost) of being in state s and taking action a



total reward $= 5r_0 + r_1$

$r_0 < 0$

Summary

Markovian Decision Process (MDP) is defined by:

Time structure: $t = 0, 1, 2, ...$

$S_t = s \in \mathcal{S}$: the state at time t

a : the action in set $\mathcal{A}(s)$

$R(s, a)$: the reward of being in state s and taking action a

$p(s'|s, a) = \mathbb{P}\{S_{t+1} = s'|S_t = s, a\}$: the prob. of transitioning to state s' if we are in state S_t and take action a

A policy is a mapping from State Space to Action Space, i.e., $\pi(s) = a$ for $s \in \mathcal{S}$

Goal is to solve for the optimal policy s.t. $\max_{\pi} U^{\pi}(s) \triangleq \mathbb{E}_{\pi}\left[\sum_{t=0}^{\infty} \rho^t \cdot R(S_t, a_t) \middle| S_0 = s\right]$

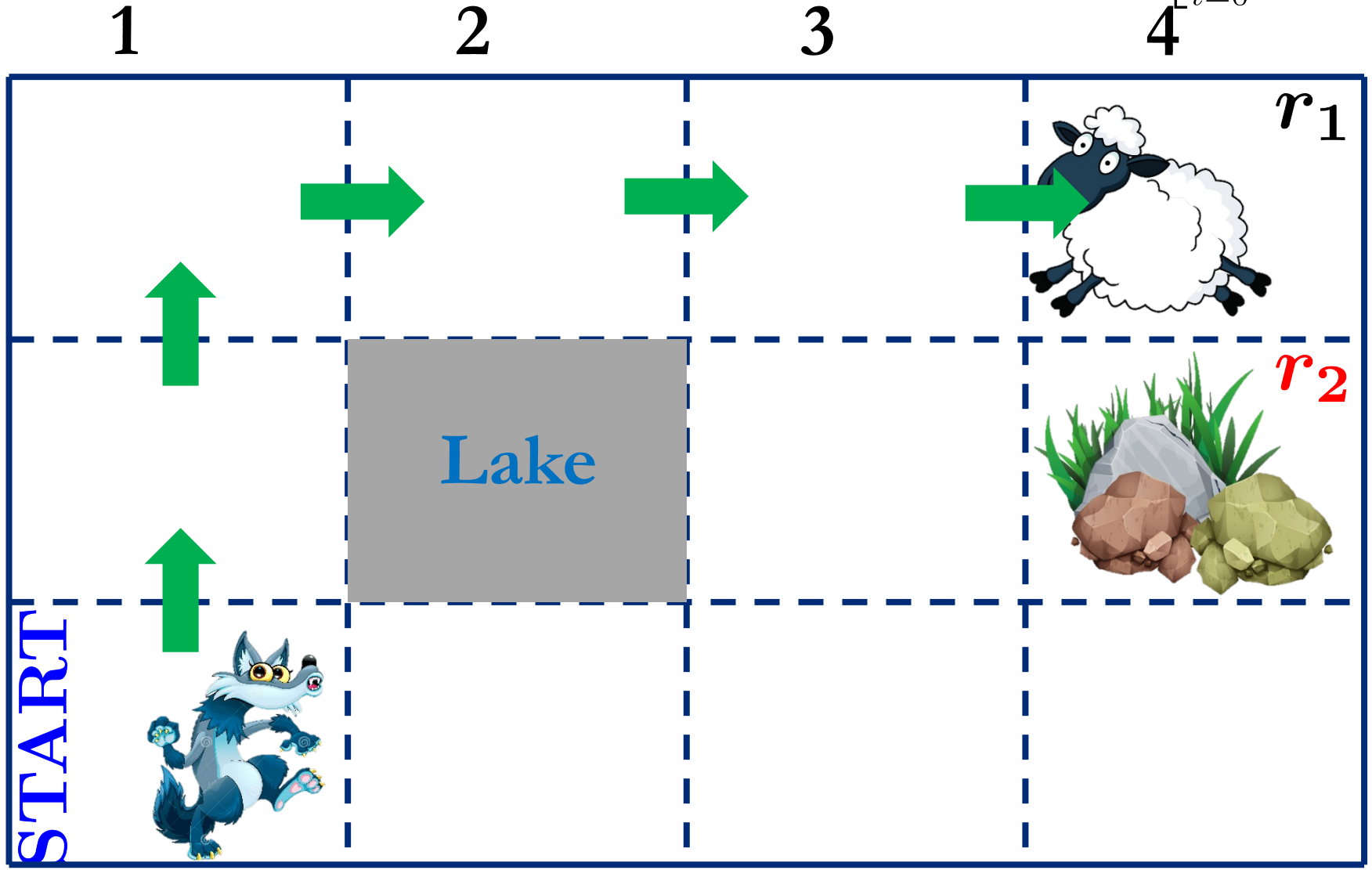
where $\rho \in (0, 1]$ is the discount factor and ρ can be 1 if it is a finite horizon problem.

$$\leq \sum_{t=0}^{\infty} \rho^t R_{\max} = \frac{R_{\max}}{1-\rho} \quad 0 < \rho < 1$$

Policy

Goal is to solve for the optimal Policy π^*

$$\max_{\pi} \mathbb{E}_{\pi}\left[\sum_{t=0}^{\infty} \rho^t \cdot R(S_t, a_t) \middle| S_0 = s\right]$$



A policy π is a function that specifies the action of the decision maker at any state.

Policy

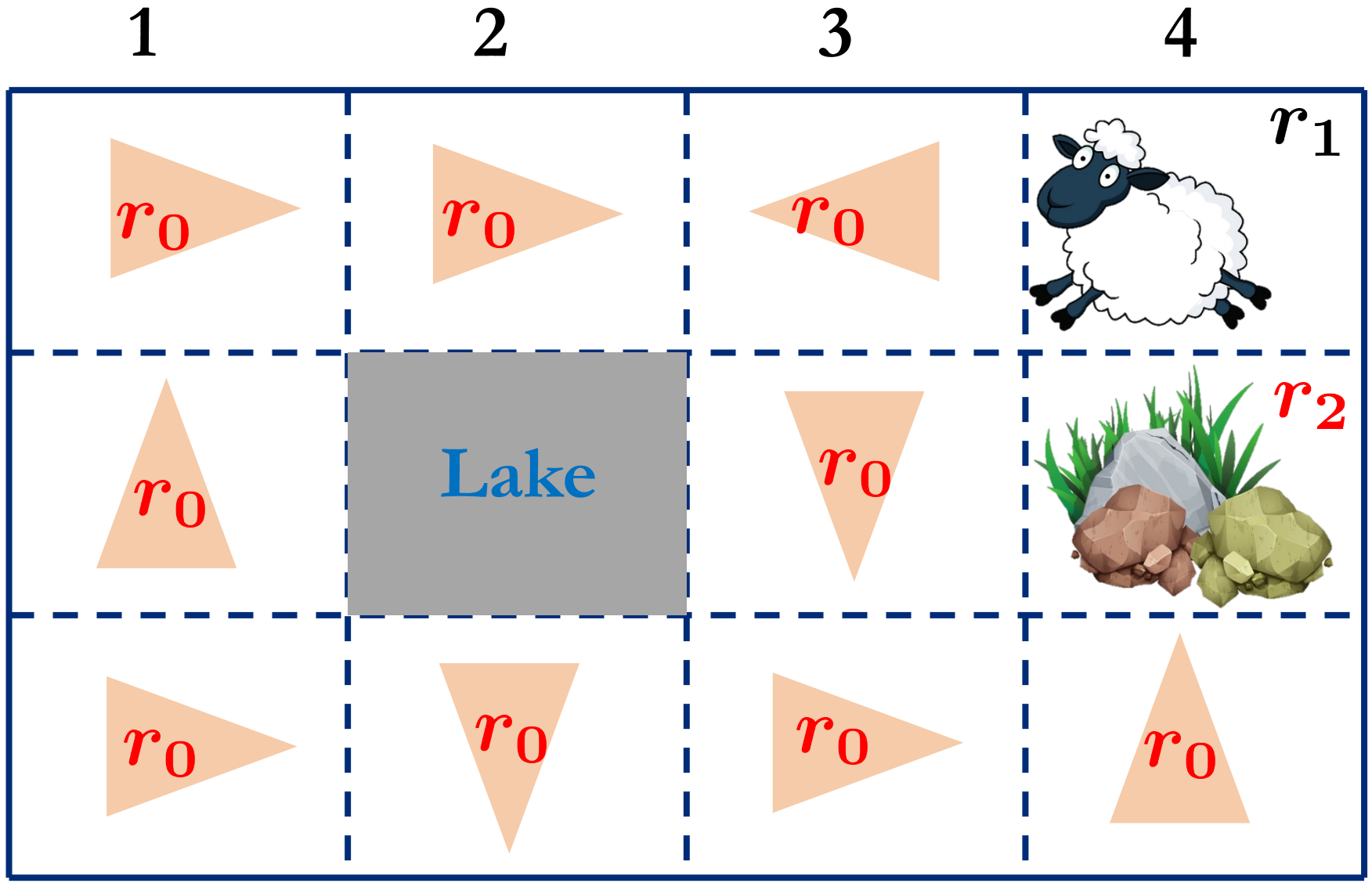
A policy π can be very complex in general (e.g., history dependent, random)

Deterministic Stationary Policy: A fixed rule that assigns one action to each state, $\pi(s) = a$

For each type of MDP, under certain conditions, the stationary policy can be optimal.

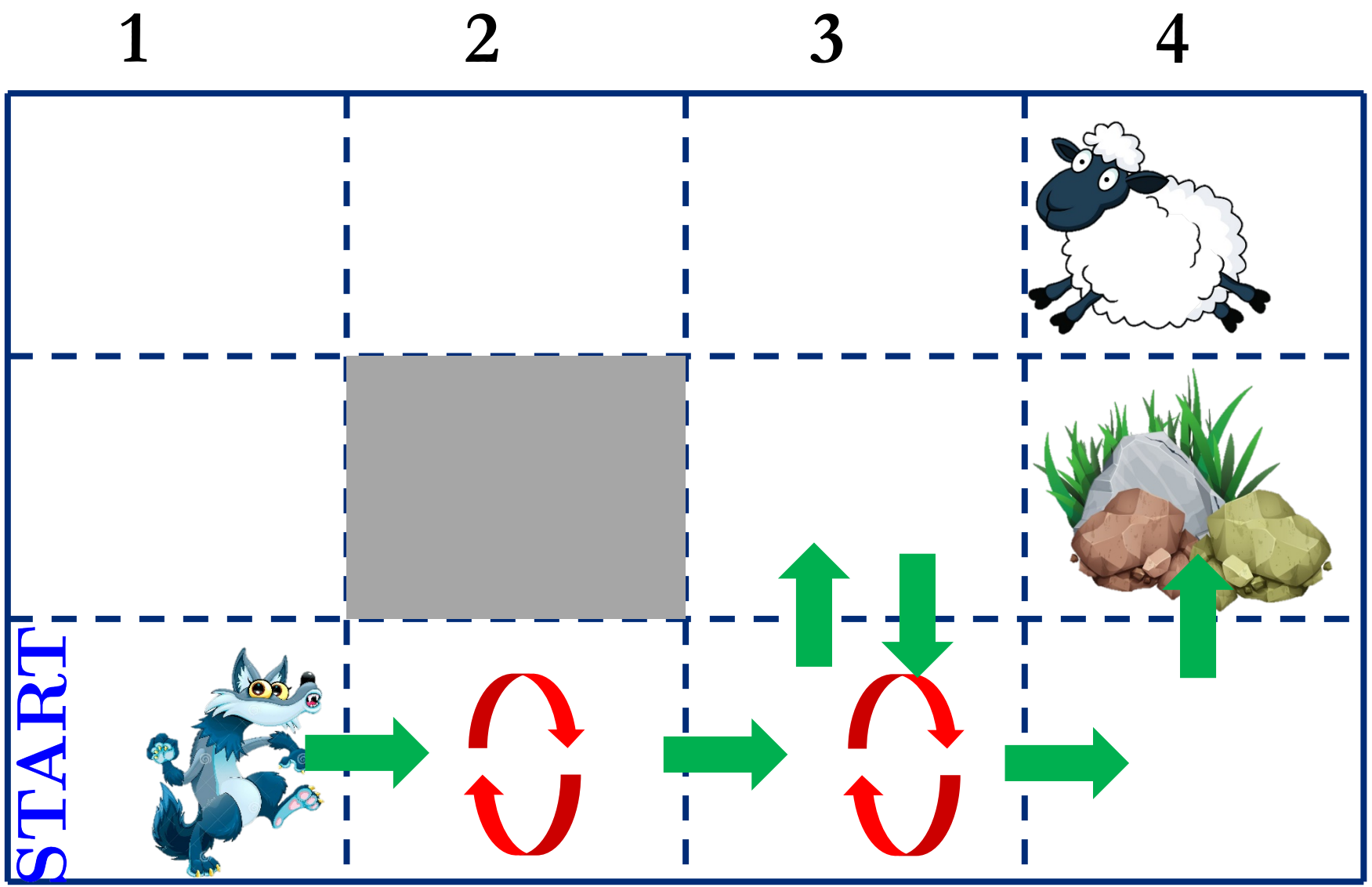
One standard result: for an infinite-horizon MDP with finite state and action spaces, bounded rewards, and $\rho \in (0,1)$, there exists an optimal deterministic stationary policy.

An Example of a Stationary Policy



$$\pi(s) = \begin{cases} \text{Up}, & \text{if } s \in \{(2,1),(3,4)\}, \\ \text{Down}, & \text{if } s \in \{(2,3),(3,2)\}, \\ \text{Right}, & \text{if } s \in \{(1,2),(3,3),(1,1),(3,1)\}, \\ \text{Left}, & \text{if } s \in \{(1,3)\}. \end{cases}$$

A Sample Path under A Stationary Policy



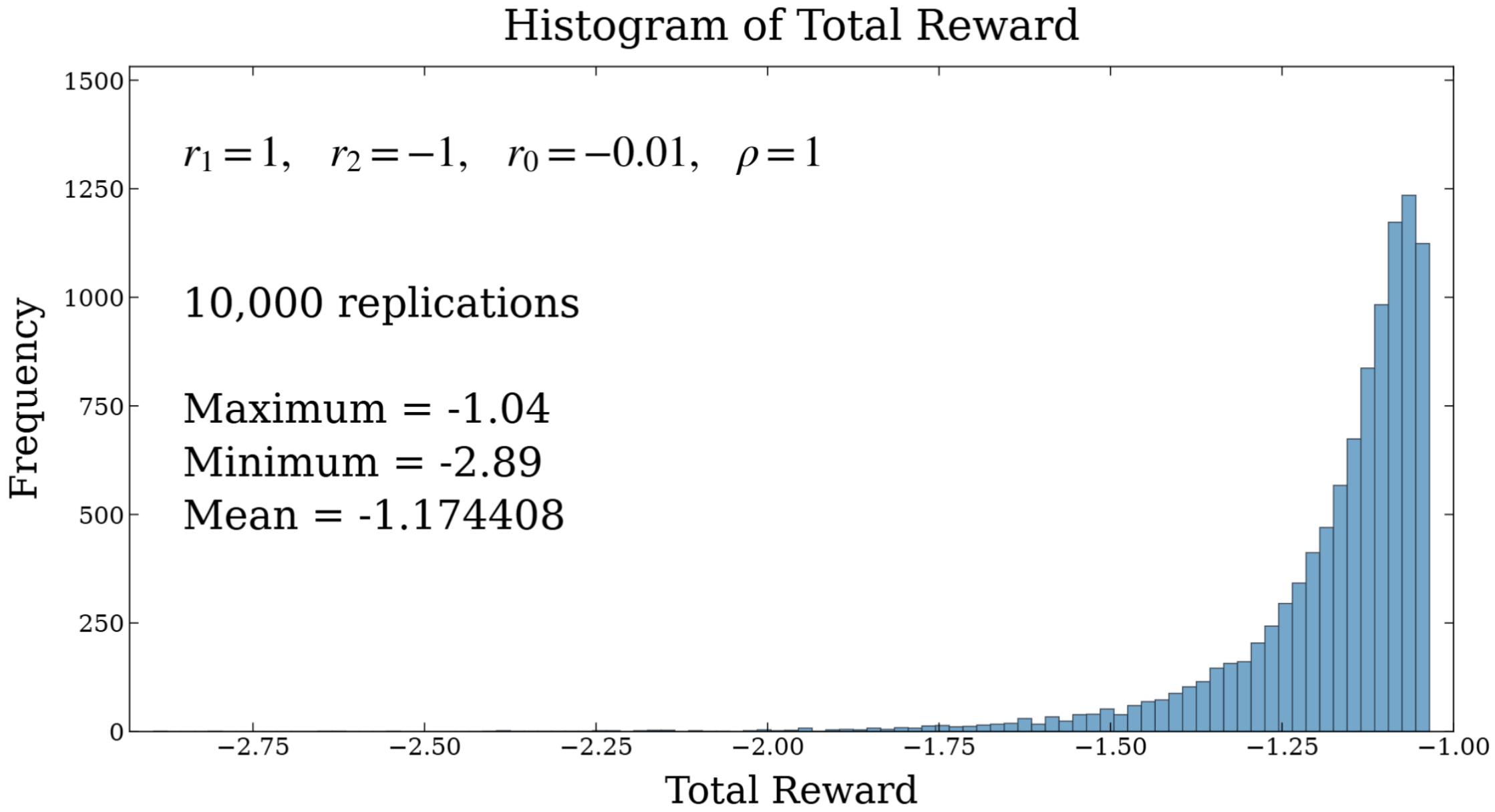
$$\pi(s) = \begin{cases} \text{Up}, & \text{if } s \in \{(2,1),(3,4)\}, \\ \text{Down}, & \text{if } s \in \{(2,3),(3,2)\}, \\ \text{Right}, & \text{if } s \in \{(1,2),(3,3),(1,1),(3,1)\}, \\ \text{Left}, & \text{if } s \in \{(1,3)\}. \end{cases}$$

Total reward of this sample path is:

$$r_0 \cdot 10 + r_2$$

$$r_0 < 0, \; r_2 < 0$$

Many Sample Paths under A Stationary Policy



All Possible Stationary Policies

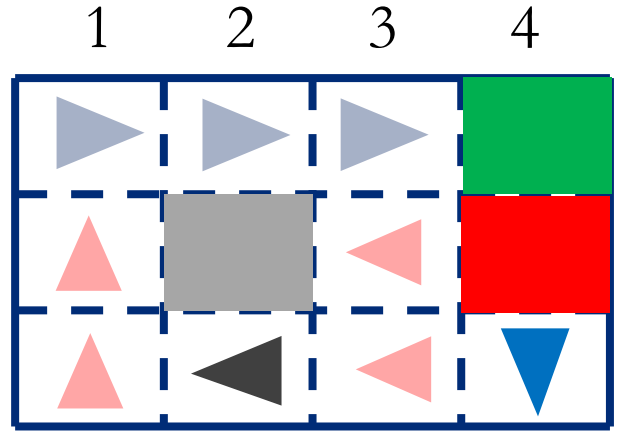
The simulation offers a way to evaluate the performance of any policy

There are totally

$$\{\text{Up, Down, Right, Left}\}^{9 \text{ states}} = 262,144$$

number of stationary policies

$$r_2 = -1$$
$$r_1 = 1$$
$$r_0 = -0.01$$
$$\rho = 1$$



I did that…

For each one of the 262,144 stationary policy, I simulate 1,000 replications and set the maximum time period to be 100. On my server with 48 cores, it took about 8 hours. The optimal policy found is

Solving MDP

Goal is to solve for the optimal policy such that:

$$U^{\pi}(s) \triangleq \mathbb{E}_{\pi}\left[\sum_{t=0}^{\infty} \rho^{t} \cdot R(S_t, a_t) \middle| S_0 = s\right]$$

$$\pi_s^* \in \arg\max_{\pi} U^{\pi}(s)$$

For the infinite-horizon problems considered here, we can choose a stationary policy π* that is optimal from every starting state. $\pi^* \equiv \pi_s^*$

Intuitively, once the wolf reaches state *c*, it faces the same future decision problem, whether it started from *d* or from *b*. We can therefore use the same optimal continuation from *c* in either case.

R(*s*,*a*) is the "short term" reward of being in state *s* and taking action a; *U*(*s*) is the "long term" expected total discounted reward from state *s* onwards under the optimal policy.

The expected utility under the optimal policy still depends on the initial state $U(s) \triangleq U^{\pi^*}(s)$

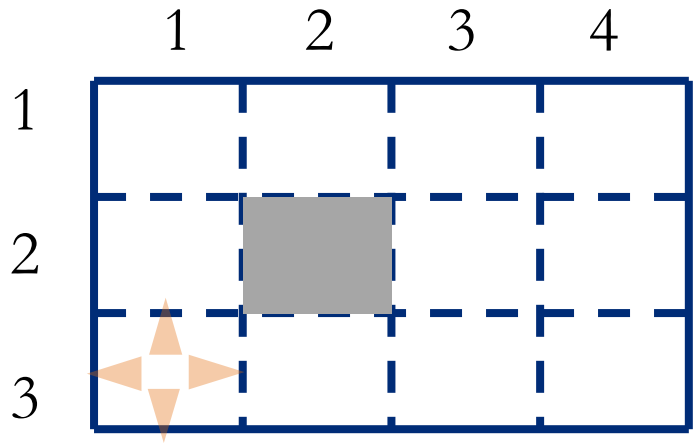
Bellman Equation

$U(s)$: the expected total discounted utility under the optimal policy given initial state s

$$U(s) = \max_{a \in \mathcal{A}(s)} \left\{ R(s,a) + \rho \sum_{s'} p(s'|s,a) U(s') \right\}$$

This is a system of non-linear equations. The number of equations equals the number of states

Example. $U(3,1)$:



$$U(3,1) = \max \left\{ r_0 + \rho \left[\begin{array}{ll} 0.8U(2,1) + 0.1U(3,2) + 0.1U(3,1), & (Up) \\ 0.9U(3,1) + 0.1U(2,1), & (Left) \\ 0.9U(3,1) + 0.1U(3,2), & (Down) \\ 0.8U(3,2) + 0.1U(2,1) + 0.1U(3,1) \end{array} \right] \right\} \quad (Right)$$

Bellman Equation

$U(s)$: the expected total discounted utility under the optimal policy given initial state s

$$U(s) = \max_{a \in \mathcal{A}(s)} \left\{ R(s,a) + \rho \sum_{s'} p(s'|s,a) U(s') \right\}$$

In general, if there are n possible states, then there are n Bellman equations, one for each state. The n equations contain n unknowns—the utilities of the states. So we would like to solve these simultaneous equations to find the $\{U(s)\}_{s \in \mathcal{S}}$

In this specific example: $U(s) = R(s) + \rho \max_{a \in \mathcal{A}(s)} \sum_{s'} p(s'|s,a) U(s')$, as the reward R is only a function of the state s

Problem: this is a system of non-linear equations

Value Iteration Algorithm

Iteration 0: Initialize $\{U_0(s)\}_{s\in\mathcal{S}}$ with arbitrary values for all non-terminal states

Set δ to be some large number

$n = 0$

While $\delta > \epsilon$ ϵ is some small number

For all non-terminal states $s \in \mathcal{S}$

$$U_{n+1}(s) \longleftarrow R(s) + \rho \max_{a\in\mathcal{A}(s)} \sum_{s'} p(s'|s,a)U_n(s')$$

$$\delta \longleftarrow \max_{s\in\mathcal{S}} |U_{n+1}(s) - U_n(s)|$$

$$n \longleftarrow n + 1$$

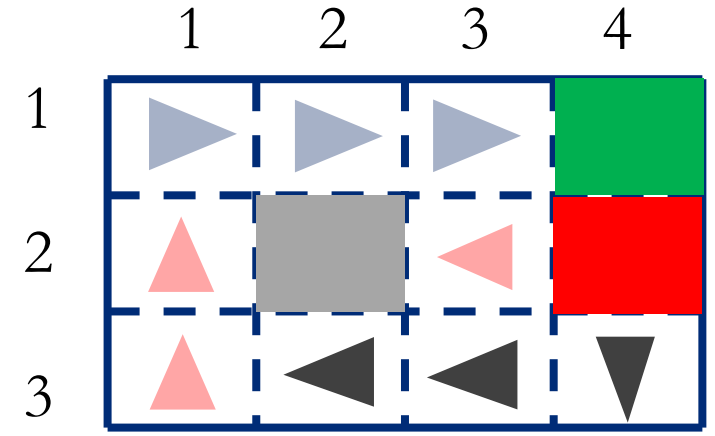
$r_2 = -1$

$r_1 = 1$

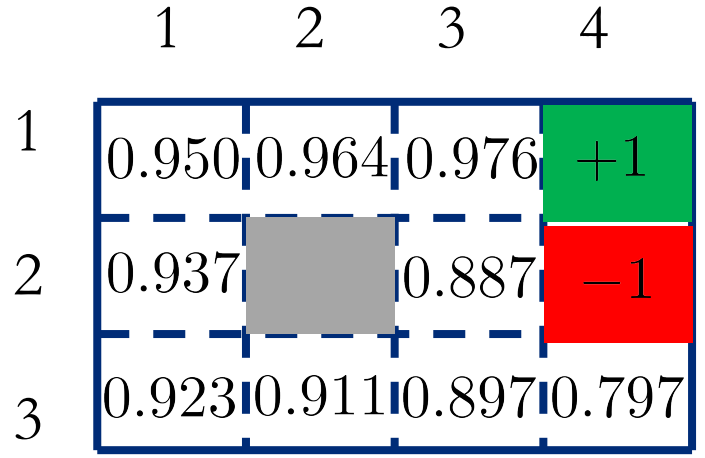
$r_0 = -0.01$

$\rho = 1$

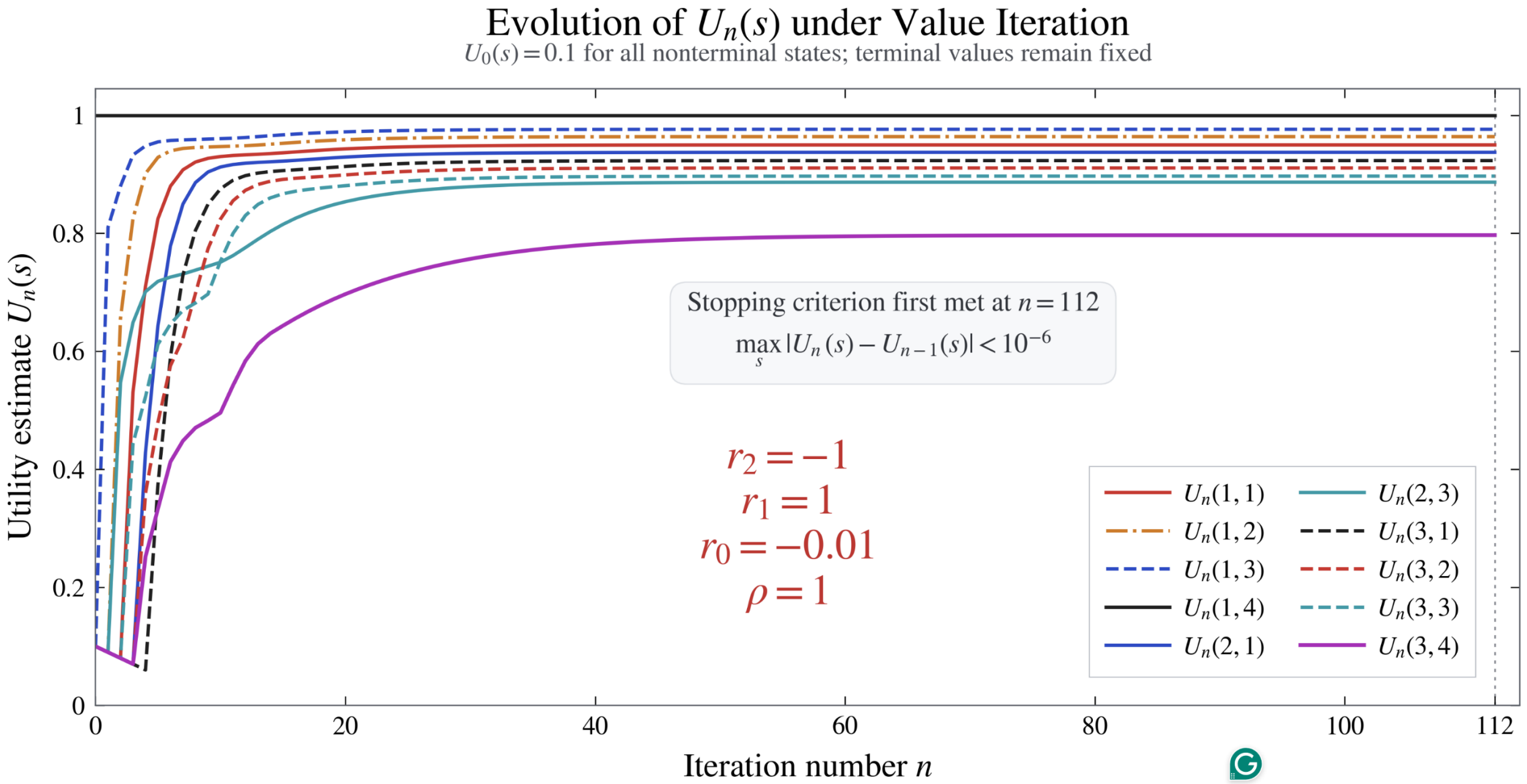
Optimal Policy π^*



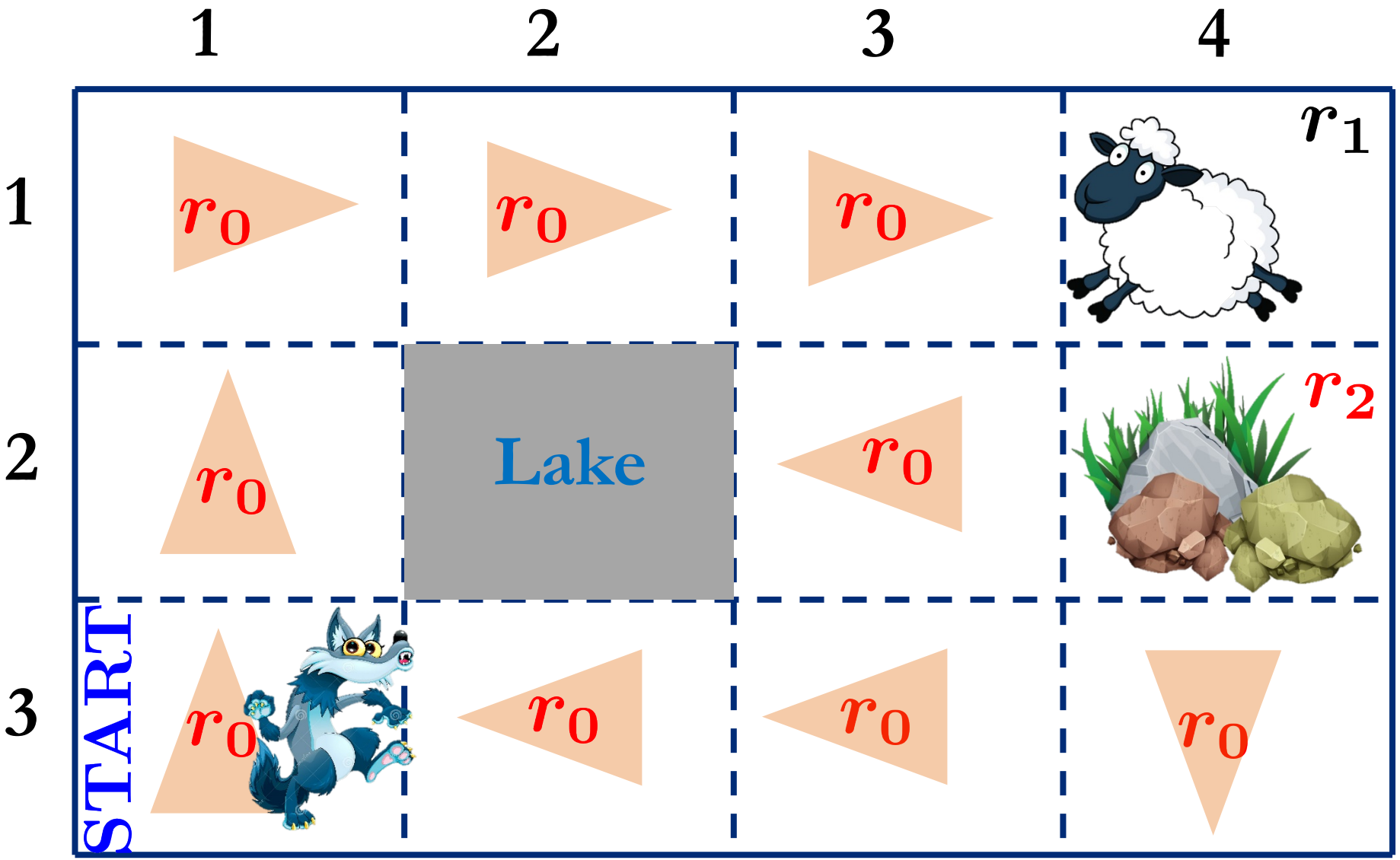
$U(s),\ \forall\ s \in \mathcal{S}$



Evolution under Value Iteration Algorithm



Other Values of the Parameters



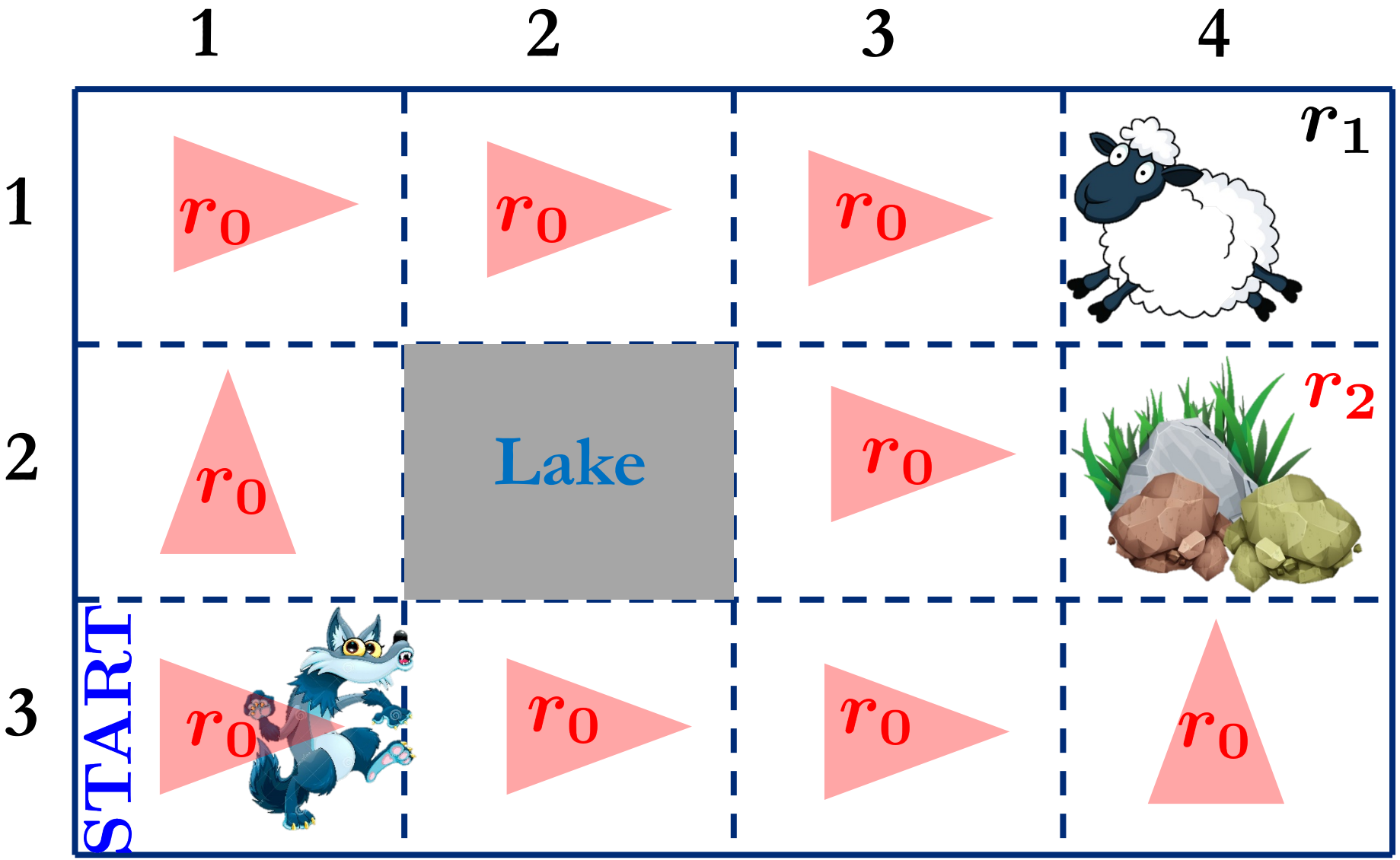
$r_2 = -1$

$r_1 = 1$

$r_0 = -0.01$

$\rho = 1$

Other Values of the Parameters



$r_2 = -1$

$r_1 = 1$

$r_0 = -1.7$

$\rho = 1$